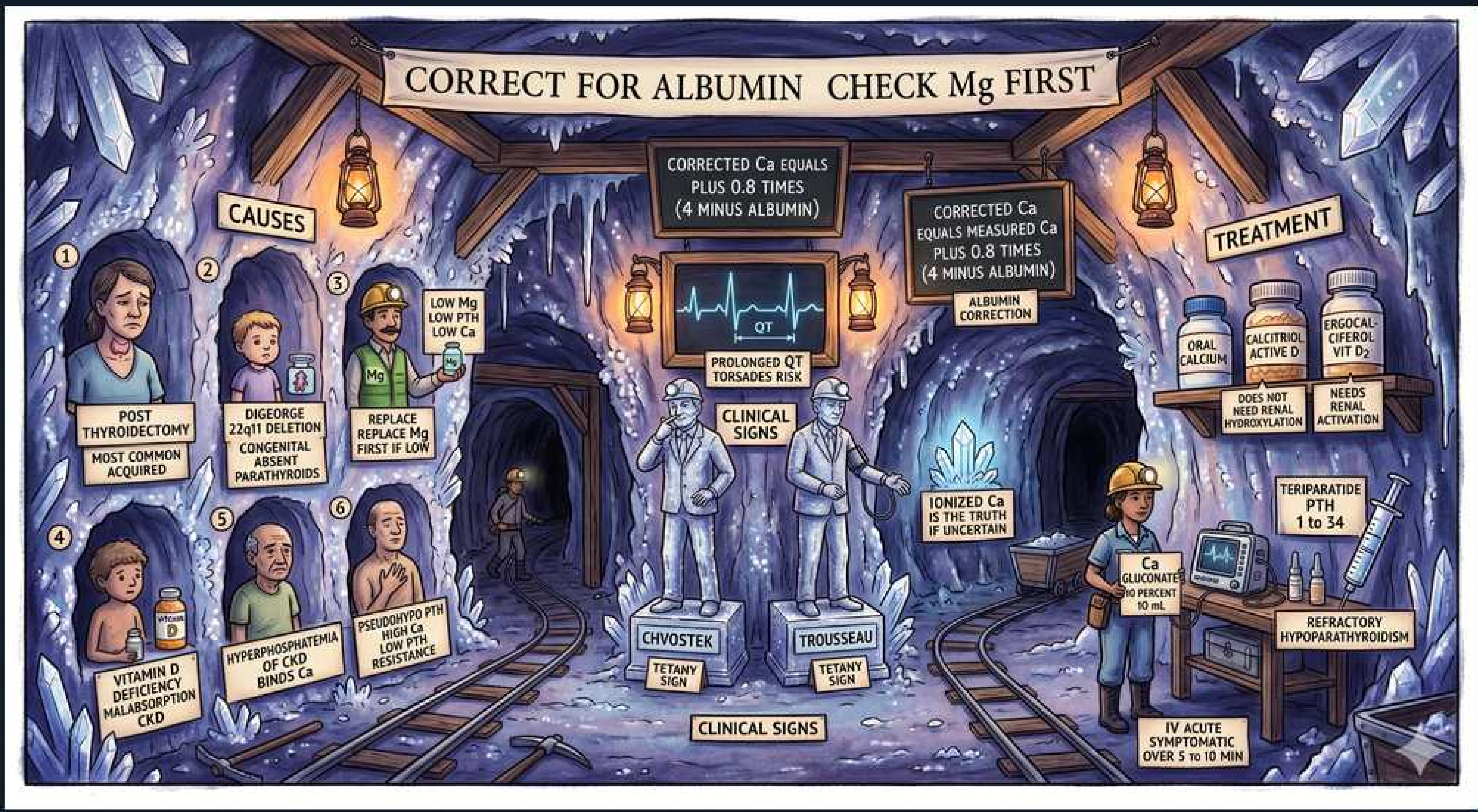
Hypocalcemia — Correct for Albumin, Check Magnesium First



1. Albumin correction formula

WHAT YOU SEE

Banner: corrected Ca = measured + 0.8 x (4 - albumin)

WHAT IT ENCODES

Total calcium falls with low albumin without true ionized deficiency. Corrected Ca = measured Ca + 0.8 x (4.0 - serum albumin). When albumin is abnormal or acid-base shifts, measure IONIZED calcium directly.

BOARD TRAP

A low total Ca in a hypoalbuminemic patient may be pseudohypocalcemia — correct or check ionized Ca before treating.

2. Check magnesium first

WHAT YOU SEE

Miner figure: low Mg replace first

WHAT IT ENCODES

Hypomagnesemia causes functional hypoparathyroidism (impaired PTH secretion AND PTH resistance) and refractory hypocalcemia. Calcium will not correct until Mg is replaced — always check and replace Mg first.

BOARD TRAP

Hypocalcemia that won't respond to calcium = check magnesium; low Mg is the classic reversible cause of refractory hypocalcemia.

3. Post-thyroidectomy hypoparathyroidism

WHAT YOU SEE

Surgical-scar figure (1)

WHAT IT ENCODES

Surgical removal/devascularization of parathyroids during thyroid or parathyroid surgery is the MOST COMMON acquired cause of hypoparathyroidism: low Ca, low PTH, high phosphate. May be transient or permanent.

BOARD TRAP

Low Ca + LOW PTH + high phosphate after neck surgery = hypoparathyroidism, not vitamin D deficiency (which has high PTH).

4. DiGeorge — 22q11 deletion

WHAT YOU SEE

Child figure, absent parathyroids (2)

WHAT IT ENCODES

DiGeorge syndrome (22q11.2 deletion) causes congenital absence of parathyroids plus thymic aplasia (T-cell immunodeficiency), conotruncal cardiac defects, and hypocalcemia. CATCH-22 mnemonic.

BOARD TRAP

Neonatal hypocalcemia + cardiac anomaly + recurrent infection = DiGeorge; contrast with Williams (hyperCALCEMIA + supravalvular AS).

5. Low Mg — replace Mg first

WHAT YOU SEE

Worker labeled low Mg (3)

WHAT IT ENCODES

Magnesium is a cofactor for PTH release; severe hypomagnesemia (PPIs, alcohol, diarrhea, aminoglycosides) blunts PTH and induces PTH resistance, yielding hypocalcemia refractory to calcium alone.

BOARD TRAP

Don't keep pushing calcium — without Mg repletion the hypocalcemia persists; this is a board-favorite reversal.

6. Vitamin D deficiency / malabsorption / CKD

WHAT YOU SEE

Vitamin D figure (4)

WHAT IT ENCODES

Low vitamin D (poor intake, malabsorption, liver disease, CKD with low 1-alpha-hydroxylase) reduces intestinal Ca absorption: low Ca, HIGH PTH (secondary hyperparathyroidism), low/normal phosphate (high in CKD).

BOARD TRAP

Vitamin D deficiency gives HIGH PTH — opposite of hypoparathyroidism; the PTH level separates the two causes of low Ca.

7. Hyperphosphatemia of CKD

WHAT YOU SEE

Figure with phosphate binders (5)

WHAT IT ENCODES

In CKD, phosphate retention binds Ca and suppresses 1,25-vitD, driving hypocalcemia and secondary hyperparathyroidism. Manage with phosphate binders, active vitamin D, and calcimimetics.

BOARD TRAP

CKD hypocalcemia comes with HIGH phosphate and HIGH PTH; treating Ca without lowering phosphate worsens vascular calcification.

8. Pseudohypoparathyroidism

WHAT YOU SEE

Figure: PTH resistance (6)

WHAT IT ENCODES

Pseudohypoparathyroidism is end-organ PTH RESISTANCE (Gs-alpha defect, Albright osteodystrophy): low Ca, high phosphate, but HIGH PTH because glands respond to the low Ca normally.

BOARD TRAP

High PTH + low Ca + high phosphate + short 4th/5th metacarpals = pseudohypoPTH (resistance), not true hypoparathyroidism (low PTH).

9. Prolonged QT / torsades

WHAT YOU SEE

ECG monitor: prolonged QT

WHAT IT ENCODES

Hypocalcemia prolongs the QT interval (specifically the ST segment), risking torsades de pointes. ECG QT prolongation is a key clinical sign and indication for urgent IV calcium.

BOARD TRAP

Prolonged QT from hypocalcemia is corrected with IV calcium — not magnesium/antiarrhythmics unless torsades is active.

---

10. Chvostek sign

WHAT YOU SEE

Facial-tap figure: tetany

WHAT IT ENCODES

Tapping the facial nerve anterior to the ear triggers facial muscle twitch — a sign of neuromuscular irritability from hypocalcemia. Sensitive but less specific (positive in some normocalcemic people).

BOARD TRAP

Chvostek is less specific than Trousseau; a positive Chvostek alone does not confirm hypocalcemia.

---

11. Trousseau sign

WHAT YOU SEE

BP-cuff figure: carpal spasm

WHAT IT ENCODES

Inflating a BP cuff above systolic for ~3 minutes induces carpopedal spasm — the MORE SPECIFIC bedside sign of hypocalcemic neuromuscular irritability.

BOARD TRAP

Trousseau (carpal spasm with cuff) is more specific than Chvostek — the tested discriminator between the two signs.

---

12. IV calcium gluconate — acute symptomatic

WHAT YOU SEE

IV calcium gluconate over 5-10 min

WHAT IT ENCODES

Symptomatic or severe hypocalcemia (tetany, seizures, QT prolongation, laryngospasm) needs IV calcium gluconate over 5-10 minutes, then an infusion. Calcium gluconate preferred peripherally (less caustic than chloride).

BOARD TRAP

Give IV calcium gluconate (not just oral) for symptomatic hypocalcemia; and replace Mg simultaneously or it won't correct.

---

13. Oral calcium + calcitriol

WHAT YOU SEE

Treatment shelf: oral Ca, calcitriol

WHAT IT ENCODES

Chronic hypoparathyroidism is treated with oral calcium plus ACTIVE vitamin D (calcitriol) because PTH-driven renal 1-alpha-hydroxylation is absent; plain vitamin D is insufficient.

BOARD TRAP

In hypoparathyroidism use calcitriol (active), not plain cholecalciferol — without PTH the kidney can't activate vitamin D.

---

14. Teriparatide (PTH 1-34) — refractory

WHAT YOU SEE

Teriparatide vial

WHAT IT ENCODES

Recombinant PTH 1-34 (teriparatide) is reserved for refractory hypoparathyroidism not controlled by calcium + calcitriol, providing the missing hormone directly.

BOARD TRAP

Teriparatide is a niche option for refractory hypoPARAthyroidism — different role than its osteoporosis use; don't confuse indications.
